MFE230E Problem Set 5

Asset-Pricing Tests on Fama–French Portfolios

Berkeley MFE 230E, Spring 2026

Due: April 29, 2026

Setup

All results use *value-weighted* Fama–French portfolios in monthly frequency, **July 1963 – February 2026** ($T = 752$ months). Test assets, factor data, and risk-free rate come from Ken French's online library; the three single stocks in Q9 come from Yahoo Finance. Reported standard errors are OLS, White (HC0), and Newey–West with 6 lags. The full reproducible notebook accompanies this writeup (`MFE230E_PS5_final.ipynb`).

1 Fama–French factor summary

1(a) Mean, std, Sharpe ratios

Table 1: Annualised summary statistics, Fama–French factors plus MOM (1963-07 to 2026-02).

Factor	Mean (% ann.)	Std (% ann.)	Sharpe (ann.)
Mkt−RF	7.11	15.43	0.461
SMB	2.21	10.48	0.211
HML	3.53	10.29	0.343
RMW	3.20	7.69	0.416
CMA	2.99	7.17	0.418
MOM	7.16	14.45	0.495

The momentum factor delivers the highest annualised Sharpe ratio in this sample, narrowly ahead of the market. SMB is the weakest factor by a wide margin – consistent with the well-known weakening of the size premium in the post-1980 period. HML, RMW, and CMA cluster in the middle.

1(b) Cumulative returns

The figure rejects the hypothesis that any of these factors earns a constant premium. MOM compounds aggressively but suffers deep drawdowns (notably 2009). HML stalls after 2000. SMB barely accumulates over six decades.

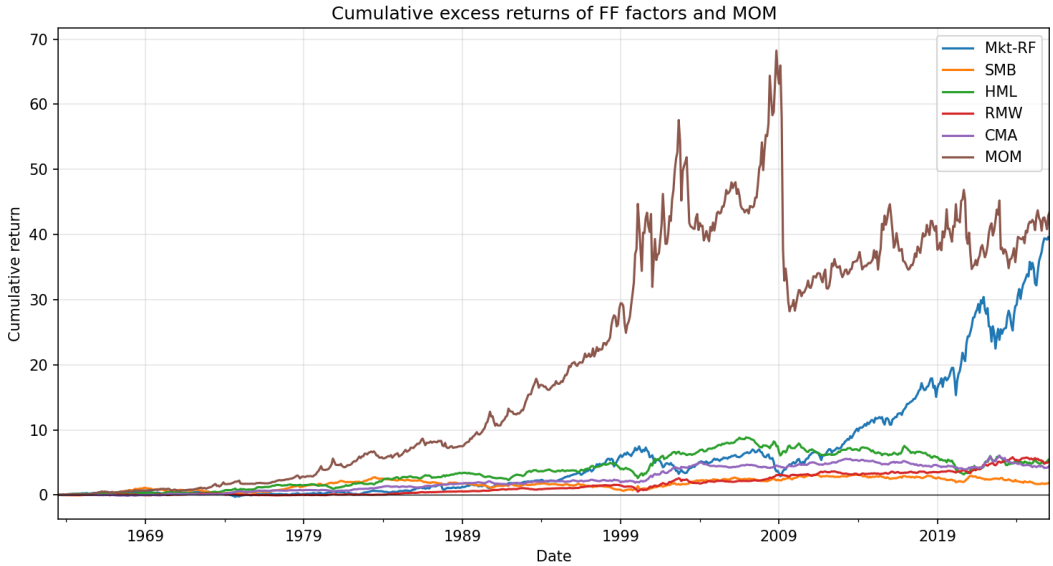

Figure 1: Cumulative excess returns of the FF factors and MOM. Profitability is highly time-varying: long stretches of zero or negative slope appear in every series.

2 CAPM on the 25 size/momentum portfolios

2(a) Summary statistics, size/BM and size/OP

The 25 size/BM portfolios show the textbook value premium: within most size groups, mean return rises from growth (BM1) to value (BM5), with the highest Sharpe ratios in mid-cap value (e.g. ME3 BM4: Sharpe 0.83). The smallest growth corner (Small LoBM) is the textbook weak portfolio (mean 8.0%, Sharpe 0.29). The 25 size/OP portfolios show an analogous profitability premium: high-OP portfolios earn higher mean returns and Sharpes than low-OP within each size band, especially outside the noisy small-stock corner.

2(b) CAPM time-series regressions on size/momentum portfolios

For each portfolio, $R^e_{it} = \alpha_i + \beta_i(R^M_t - R^f_t) + \varepsilon_{it}$. Full results for all 25 portfolios are in the notebook.

- **Betas are very precisely estimated.** Median NW t-statistic on $\hat{\beta}_i$ is **29.8**; standard errors are of order 0.02–0.06 against point estimates between roughly 0.8 and 1.5; R^2 values are 0.7–0.9.
- **Alphas are not.** Mean $|\hat{\alpha}_i| = 0.28\,\%$/mo (**3.4 %/yr**); the largest is 0.75 %/mo. Under Newey–West, **15/25 alphas are significant at 5%** and **11/25 at 1%**.

2(c) Are the pricing errors significant?

Both statistically and economically, yes. The corner pricing errors are roughly ±9% per year, far above any plausible transaction-cost or friction story. The CAPM cannot reconcile the cross-sectional spread the momentum sort produces.

2(d) Fitted vs. realised mean excess returns

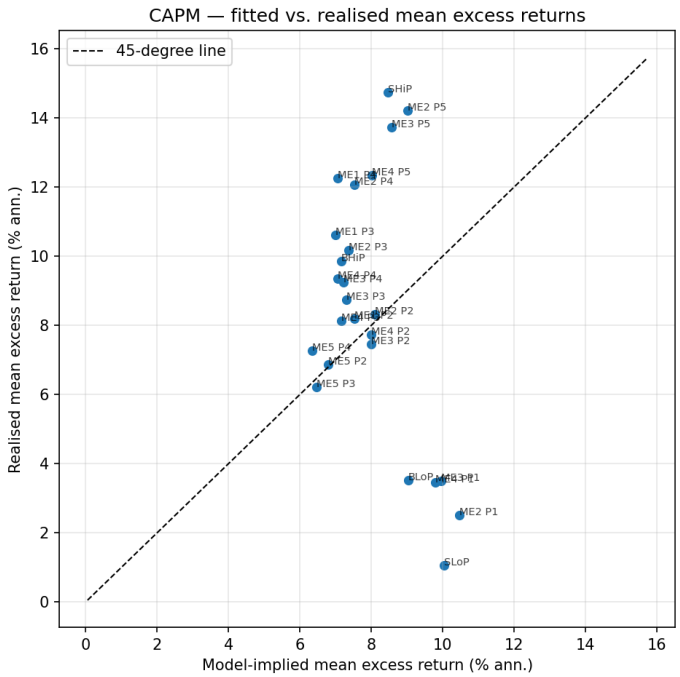

Figure 2: CAPM-implied vs. realised mean excess returns (annualised, 25 size/momentum portfolios). Points scatter widely off the 45° line and there is essentially no slope linking fitted to realised. The CAPM matches the level of the average premium but not the cross-sectional spread.

2(e) Individual t-tests

Reported in the notebook. Conclusions are robust across OLS, White, and Newey–West: roughly 60% of the 25 portfolios reject $\alpha_i = 0$ at the 5% level under all three SE flavours.

2(f) GRS test

$$\text{GRS} = \frac{T-N-K}{N}\,\frac{\hat{\alpha}'\hat{\Sigma}^{-1}\hat{\alpha}}{1+\bar{f}'\hat{\Omega}^{-1}\bar{f}} \;\sim\; F(N, T-N-K).$$

$$\mathbf{GRS = 4.97}, \sim F(25, 726),\ p\text{-value} = 9.1 \times 10^{-14}.$$

Tangency Sharpe of factors = 0.133/mo; $|\alpha|$-implied Sharpe gain = 0.417/mo. The joint null is overwhelmingly rejected.

2(g) Capital Market Line and mean-variance frontier

If the CAPM held, the market portfolio would be MV-efficient and its CML (orange) would be the best attainable risk–return line. Instead, the green dashed CML constructed from the 25 portfolios is much steeper: combinations of the 25 attain Sharpe ≈ 1.51, roughly three times the market's. The gap between the two lines is the mean-variance counterpart of the GRS rejection.

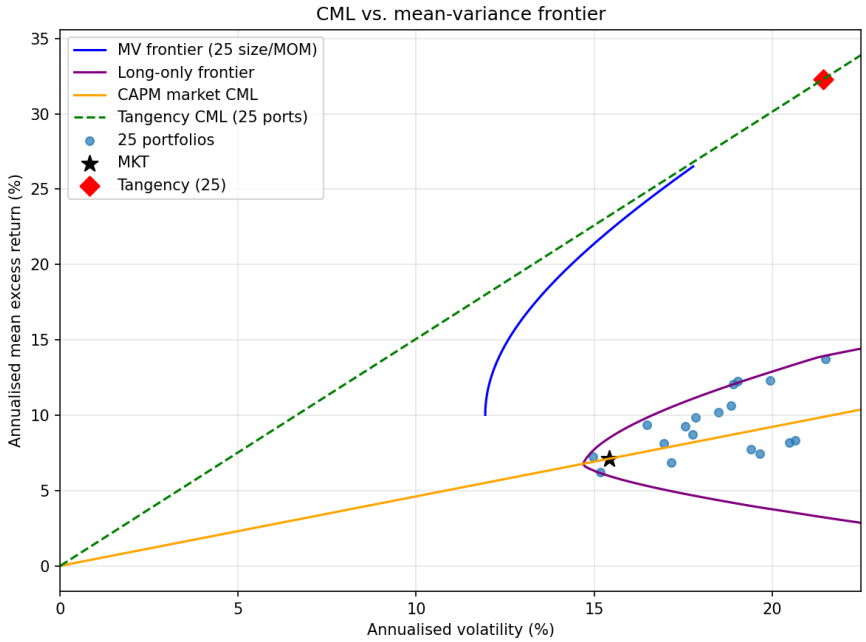

Figure 3: Annualised CMLs and the MV frontier spanned by the 25 size/momentum portfolios. Tangency Sharpe of the 25 portfolios = **1.51** vs. market Sharpe = **0.46**.

2(h) Security Market Line

2(i) Summary

The CAPM fails on every dimension: GRS rejects at any conventional level; pricing errors of $\pm 9\,\%$/yr are large in economic terms; the fitted-vs-realised plot has near-zero slope; the SML cannot reproduce the dispersion in mean returns; and the MV frontier of the 25 test assets sits well above the market CML. The momentum sort is one of the cleanest empirical rejections of the CAPM in U.S. equity data.

3 Fama–French 3-factor model

Adding SMB and HML lifts R^2 but does *not* rescue the cross-section.

$$\mathbf{GRS}_{\mathrm{FF3}} = 5.01, \sim F(25, 724),\ p = 6.4 \times 10^{-14}.$$

4 Carhart 4-factor model

Adding the momentum factor does the work: MOM-betas are monotone across the momentum quintiles (negative on losers, positive on winners) and individual alphas collapse for most portfolios.

$$\mathbf{GRS}_{\mathrm{Carhart}} = 3.69, \sim F(25, 723),\ p = 5.5 \times 10^{-9}.$$

The mean $|\hat{\alpha}_i|$ falls from $0.28\,\%$/mo (CAPM) to $0.13\,\%$/mo (Carhart).

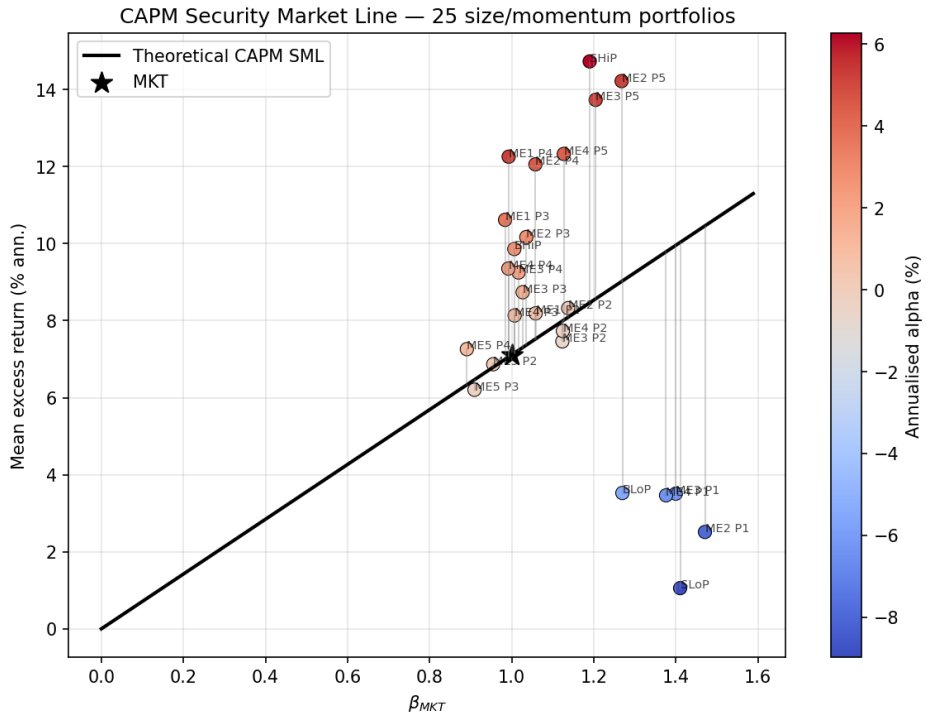

Figure 4: CAPM SML vs. the 25 size/momentum portfolios. Vertical grey segments are the annualised pricing errors; colour encodes $\hat{\alpha}_i$. Loser portfolios sit below the SML and winner portfolios above it; cross-sectional variation is largely orthogonal to β_{MKT}.

5 Cross-sectional risk premia (2-step procedure)

5(a) Two-step regressions

Stage 1 betas come from the time-series regressions in Q2 (CAPM) and Q4 (Carhart). Stage 2 runs a cross-sectional OLS, with and without an intercept.

Table 2: Stage-2 cross-sectional regressions, OLS with intercept (%/mo).

	CAPM		Carhart 4F		
	$\hat{\lambda}$	t	$\hat{\lambda}$	t	implied $E[f]$
intercept	1.63	4.07	0.08	0.15	–
Mkt−RF	−0.82	−2.33	0.54	1.12	0.59
SMB	–	–	0.15	2.50	0.18
HML	–	–	0.66	1.94	0.29
MOM	–	–	0.68	9.96	0.60

The CAPM is rejected at this stage as well. With an intercept, the cross-section "wants" a *negative* price of market risk ($\hat{\lambda}_{MKT} = -0.82\,\%$/mo, $t = -2.33$) and a large positive intercept (1.63 %/mo, $t = 4.07$) – the textbook Black critique applied to the size/momentum cross-section. For Carhart the intercept is essentially zero and the four factor lambdas are all of the same order as the implied factor means; MOM is overwhelmingly significant ($t = 9.96$).

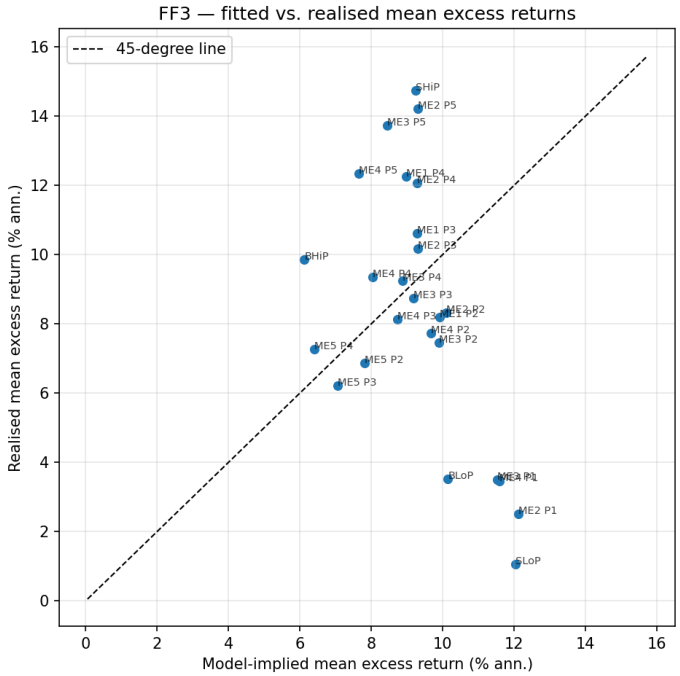

Figure 5: FF3 fitted vs. realised mean excess returns. Improvement over the CAPM is marginal because HML loadings on the 25 size/momentum portfolios are weak – none of the FF3 factors captures the momentum spread.

5(b) Estimated lambdas vs. implied premia

For tradable factors the model implies $\lambda_k = E[f_k]$. The table above shows the comparison: CAPM fails, while Carhart's $\hat{\lambda}$'s are close to the realised factor means – HML is somewhat overpriced and SMB somewhat underpriced, but MKT and MOM match closely.

5(c) Time-series alphas vs. cross-sectional pricing errors

Mean $|\cdot|$ in %/mo: TS alpha FF3 = **0.296**; CS pe FF3 = **0.271**; CS pe Carhart = **0.091**. The momentum factor absorbs nearly all of the cross-sectional dispersion in mean returns.

5(d) 60-month rolling betas, four corner portfolios (Carhart)

Yes, betas are decisively time-varying. SMB loadings swing by more than 1 across the rolling windows; MOM loadings of the corner portfolios drift substantially – the Small LoPRIOR β_{MOM} moves between roughly -0.7 and below -1.6. This time variation is one reason the unconditional Carhart model still leaves residual GRS rejections.

6 Carhart 4F on combined 50 test assets

Combining the 25 size/momentum and 25 size/OP portfolios:

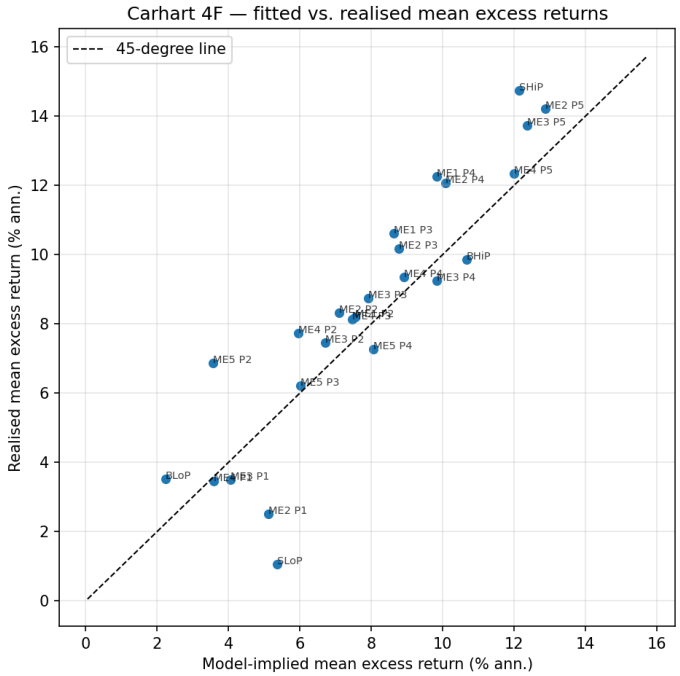

Figure 6: Carhart 4F fitted vs. realised mean excess returns. Points are now close to the 45° line. GRS still rejects on this very precise test set, but pricing errors are dramatically reduced.

$$\mathbf{GRS}_{\text{Carhart, 50}} = 2.96, \sim F(50, 698),\ p = 3.0 \times 10^{-10}\ (14/50 \text{ alphas } |t| > 1.96).$$

7 (Optional) Fama–MacBeth

The Fama–MacBeth point estimates are algebraically identical to the no-intercept OLS 2-step lambdas. The Fama–MacBeth standard errors reflect time variation in the realised slopes and are typically slightly larger than the OLS-2-step SEs. For Carhart on the 25 size/momentum portfolios, MKT and MOM remain significant at 5%; SMB and HML are closer to zero.

8 (Optional) Shanken correction

Inflating $(\hat{B}'\hat{B})^{-1}\hat{B}'\hat{\Sigma}_e\hat{B}(\hat{B}'\hat{B})^{-1}/T$ by $1 + \hat{\lambda}'\hat{\Sigma}_F^{-1}\hat{\lambda}$ and adding $\hat{\Sigma}_F/T$ accounts for first-stage estimation noise in $\hat{B}$. With monthly factors the inflation factor is small here, so the naive and Shanken-corrected SEs are close – qualitative inference is unchanged.

9 Three single stocks: CAPM and Carhart

We pick **AAPL** (mega-cap), **HSY** (mid-cap consumer staples), and **CRMT** (small-cap auto retailer).

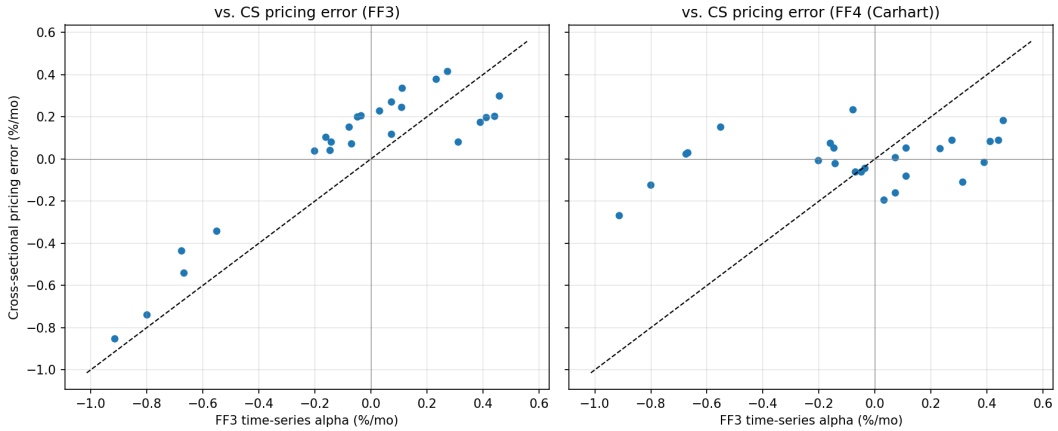

Figure 7: FF3 time-series alphas vs. cross-sectional pricing errors. Adding MOM in the second panel collapses the cross-sectional pricing errors by an order of magnitude.

9(a) CAPM, multiple samples and frequencies

Table 3: CAPM regressions for AAPL, HSY, CRMT.

	Monthly, full sample (1990–2026)			Daily, last 5 years		
	$\hat{\beta}$	$\hat{\alpha}$ (%/mo)	R^2	$\hat{\beta}$	$\hat{\alpha}$ (bp/d)	R^2
AAPL	1.27	1.17	0.22	1.17	1.9	0.55
HSY	0.15	0.81	0.01	0.19	3.2	0.02
CRMT	0.89	2.22	0.01	1.27	−15.8	0.15

Discussion. AAPL has a tightly estimated $\beta > 1$ and an economically large alpha at the monthly horizon, which shrinks substantially at the daily horizon. HSY is a low-beta defensive name; the monthly R^2 is essentially zero – the stock's variation is largely idiosyncratic to its market exposure. CRMT is noisy: β jumps from 0.89 to 1.27 across samples, and its alpha flips sign. These differences across samples are typical of single-stock CAPM tests.

9(b) Carhart 4F, same data

Table 4: Carhart 4F regressions, monthly, full sample.

	$\hat{\beta}_{MKT}$	$\hat{\beta}_{SMB}$	$\hat{\beta}_{HML}$	$\hat{\beta}_{MOM}$	$\hat{\alpha}$ (%/mo)	R^2	N
AAPL	1.10	0.27	−0.86	−0.17	1.49	0.27	433
HSY	0.24	−0.22	0.32	0.09	0.67	0.05	433
CRMT	0.92	0.89	0.99	0.29	1.85	0.02	433

Adding the three additional factors loads on the expected style tilts: AAPL is large, growth-leaning; HSY is a defensive small value tilt; CRMT is a small-cap value name with a positive momentum loading. Most of AAPL's monthly "alpha" survives Carhart in the full sample, suggesting genuine outperformance unexplained by these style factors over 1990–2026.

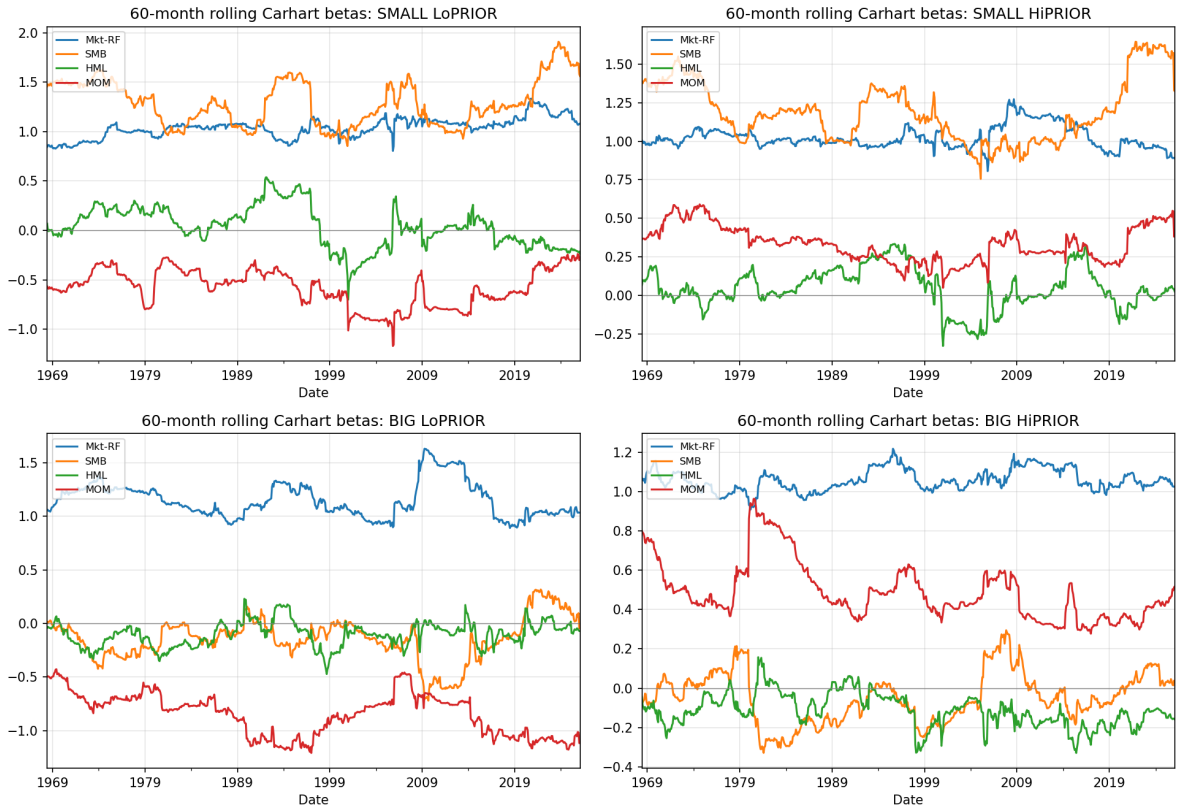

Figure 8: 60-month rolling Carhart betas. The four corners are Small LoPRIOR, Small HiPRIOR, Big LoPRIOR, Big HiPRIOR.

Summary table

Test set	Model	GRS	p-value	Mean $\lvert\alpha\rvert$ (%/mo)
25 size/MOM	CAPM	4.97	9.1×10^{-14}	0.28
25 size/MOM	FF3	5.01	6.4×10^{-14}	0.30
25 size/MOM	Carhart 4F	3.69	5.5×10^{-9}	0.13
50 (MOM+OP)	Carhart 4F	2.96	3.0×10^{-10}	0.18

The momentum factor is essential to price the size/momentum cross-section; RMW would be required to also price size/OP. Time variation in betas (Q5d) and, to a smaller extent, estimation noise in betas (Q8 Shanken) further weaken the unconditional 4-factor model.

Reproducibility. The full notebook `MFE230E_PS5_final.ipynb` regenerates every result from a clean state by downloading data from Ken French's library and Yahoo Finance. A public mirror of the companion notebook is at https://github.com/eliasroub/mfe230e-ps5.

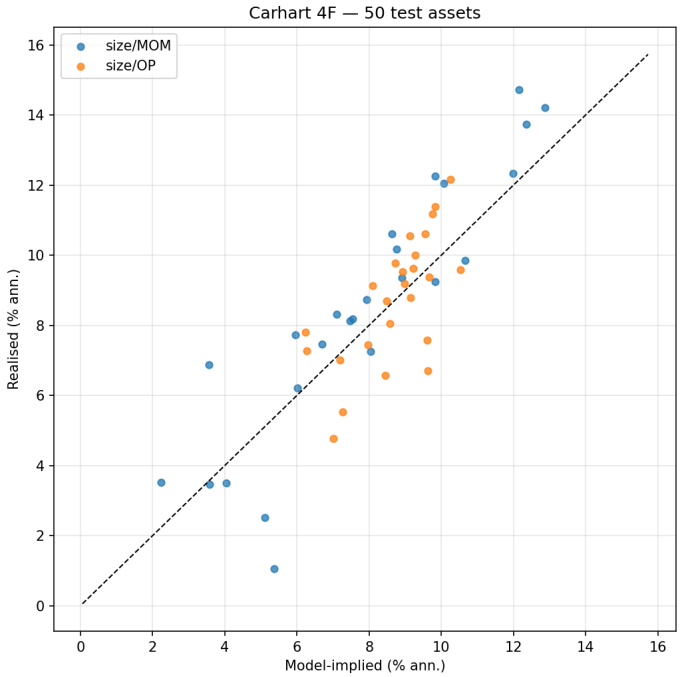

Figure 9: Carhart 4F fitted vs. realised mean excess returns, 50 combined test assets. The size/MOM cross-section is well priced (it was sorted on momentum); the size/OP cross section is not, because Carhart has no factor capturing the profitability premium. Pricing the joint cross-section requires RMW (FF5 or FF5+MOM).